Project proposal

# Lost in Subtitles

Detecting translation errors in English to Hebrew subtitles with LLMs

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Large Language Models and Their Applications · Bar-Ilan University



Motivating use case

# Israelis watch most of their TV in English. The subtitles carry the meaning.

**Real example · The Wire, Hebrew subtitles**

“Who gives a rat's ass?” means “who cares”. In Hebrew it came out as:

לאיזה תחת לתקוע עכברוש

## Why it matters

A bad subtitle kills the joke, flips the meaning or drops half the line. Streaming services are moving to machine translation, so these errors are getting more common.

## Why it's hard

The errors hide in idioms, humor, sarcasm and tone. Catching them needs both languages and the context. There is no labeled English–Hebrew dataset of subtitle errors.

## How it's handled today

Human reviewers check subtitles by hand. Automatic quality scores give one number, not what went wrong.

# Task description

Input
**English line + Hebrew subtitle**

→ **Classifier** →

Output
**Correct, or error type**

| Label | English | Hebrew subtitle |
|---|---|---|
| Correct | I'm not going anywhere | אני לא הולך לשום מקום |
| Meaning error | You can't miss it | אסור לך לפספס את זה |
| Literal idiom | Break a leg | תשבור רגל |
| Omission | Sorry, I didn't mean it, I swear | סליחה |
| Wrong tone | Would you mind closing the door? | סגור את הדלת |

## What's new

### Hebrew

First labeled dataset of English to Hebrew subtitle errors

### What went wrong

An error type, not just a quality score

### Synthetic to real

Trained on injected errors, tested on real ones

# Models & methods

**1 · Collect**
Real English–Hebrew line pairs from OpenSubtitles

**2 · Inject**
An LLM adds one error of a chosen type

**3 · Train & prompt**
Five methods on the same data

**4 · Evaluate**
Synthetic test set and a real test set

| Method | Kind | Training |
|---|---|---|
| **LLM zero-shot** (baseline) | Off-the-shelf | None · model TBD |
| **LLM + system prompt** with an error guide | Off-the-shelf | None |
| **LLM few-shot** with solved examples | Off-the-shelf | None |
| **Multilingual encoder** (e.g. XLM-R) | Fine-tuned | Full fine-tune on our pairs |
| **Small LLM + LoRA** | Fine-tuned | LoRA · model TBD |

# Data

### Correct pairs · real

**Source:** OpenSubtitles (OPUS), English–Hebrew aligned lines

**Focus:** comedies and sitcoms, where idioms and humor live. Filtering by title: TBD

**Size:** ~6,000 pairs in total, about 750 per error type

### Error pairs · synthetic

**How:** an LLM gets a correct pair and an error type, and rewrites the Hebrew line with exactly that error

**Balance:** half correct, half errors, spread over four types

**Generators:** model(s) TBD

### Labels

Known by construction: the error we asked for is the label

### Quality check

Manual review of a random sample: is there exactly one real error?

### Real test set

~150 real pairs labeled by hand, never used for training

# Metrics & KPIs

### Detection

F1 and AUC: correct vs. error

### Error type

Macro F1 over types, plus a confusion matrix

### Synthetic to real

The same metrics on the hand-labeled real set

### Data quality

Share of injected errors that pass manual review

| Method | Detect F1 | Type F1 | Real-set F1 |
|---|---|---|---|
| LLM zero-shot | – | – | – |
| LLM + system prompt | – | – | – |
| LLM few-shot | – | – | – |
| Multilingual encoder | – | – | – |
| Small LLM + LoRA | – | – | – |

Filled in the interim and final reports. Ground truth: the injected error type, and hand labels on the real set.

# Hard cases & limits

Game of Thrones · S6E5 · The Door

English

**Hold the door! → Hodor**

Hebrew subtitle

תחזיק את הדלת

Every word is right. The meaning is gone.

## What we catch

Errors inside the line itself: wrong meaning, literal idioms, missing parts, wrong tone.

## What stays out

Wordplay that depends on context outside the line, like Hodor. The model sees one line, not six seasons.

## Future work

Add the previous lines of the scene as input.